

Nested Loops

Learning Objectives

- Construct nested loops and trace their execution step by step
- Apply nested loops to process multi-dimensional data patterns
- Analyze the time complexity of nested loop structures

Engage (5-7 minutes)

A multiplication table requires row-and-column thinking-each cell is the product of its row and column numbers. A single loop handles one dimension, but nested loops handle two or more dimensions. From printing patterns to processing matrices, nested loops are everywhere. Today we learn to think in multiple dimensions and understand how loop nesting affects performance.

Explore (10-12 minutes)

Write a nested loop that prints a 5x5 grid of asterisks. Then modify it to print a right triangle where row 1 has one star, row 2 has two stars, and so on. Count how many times the inner loop body executes for each pattern. Now try a 10x10 grid-how many total iterations does that produce? Discuss with a partner why nested loops can be slow on large datasets.

Explain (10-12 minutes)

A nested loop is a loop inside another loop. The outer loop controls the rows or high-level iterations, while the inner loop handles columns or sub-iterations within each outer step. For an outer loop running N times and an inner loop running M times, the total iterations are $N \times M$, making the time complexity $O(N \times M)$. Nested loops are essential for processing 2D arrays, generating patterns, and implementing algorithms like matrix multiplication.

Elaborate (8-10 minutes)

Write nested loops to: print a diamond pattern of stars, transpose a 4x4 matrix (swap rows and columns), and find all pairs of numbers in two arrays that sum to a target value. Then implement a selection sort algorithm using nested loops and count the comparisons. Discuss: when would you use three levels of nesting, and what performance concerns arise?

Evaluate (5-8 minutes)

Trace through this nested loop and list every output: outer loop i from 1 to 4, inner loop j from 1 to i , print $i*j$. Then write a nested loop that prints a hollow square pattern of size 6. Explain why a nested loop with 1000 iterations in both loops produces 1,000,000 total executions.